

Minimal Constraint Subsets found in 11-round trail~Zhang's paper Figure2 Skinny64-192

the 0 th Minimal Constraint set is:

$$\begin{aligned}
 &y^2_{2_1} + x^3_{5_1} + k_{1_7} + k_{2_7_1} + k_{3_7_1} = 0 \\
 &y^2_{2_1} + y^2_{11_1} + x^3_{13_1} + k_{1_7} + k_{2_7_1} + k_{3_7_1} = 0 \\
 &[y^2_{4_1}] + y^2_{11_1} + [x^3_{9_1}] + k_{1_2} + k_{2_2_1} + k_{3_2_1} = 0 \\
 &[y^3_{0_1}] + y^3_{10_1} + y^3_{13_1} + [x^4_{0_1}] + k_{1_{15}} + k_{2_{15_1}} + k_{3_{15_1}} = 0 \\
 &y^3_{2_1} + x^4_{6_1} + k_{1_9} + k_{2_9_1} + k_{3_9_1} = 0 \\
 &y^3_{2_1} + y^3_{8_1} + x^4_{14_1} + k_{1_9} + k_{2_9_1} + k_{3_9_1} = 0 \\
 &y^3_{5_1} + y^3_{8_1} + x^4_{10_1} + k_{1_{12}} + k_{2_{12_1}} + k_{3_{12_1}} = 0 \\
 &[y^3_{7_1}] + y^3_{10_1} + [x^4_{8_1}] + k_{1_{13}} + k_{2_{13_1}} + k_{3_{13_1}} = 0 \\
 &[y^4_{0_1}] + y^4_{10_1} + [y^4_{13_1}] + x^5_{0_1} + k_{1_7} + k_{2_7_2} + k_{3_7_2} = 0 \\
 &[y^4_{1_1}] + [y^4_{11_1}] + y^4_{14_1} + x^5_{1_1} + k_{1_3} + k_{2_3_2} + k_{3_3_2} = 0 \\
 &[y^4_{2_1}] + [y^4_{8_1}] + x^5_{14_1} + k_{1_1} + k_{2_1_2} + k_{3_1_2} = 0 \\
 &[y^4_{3_1}] + y^4_{9_1} + [x^5_{15_1}] + k_{1_6} + k_{2_6_2} + k_{3_6_2} = 0 \\
 &y^4_{6_1} + y^4_{9_1} + x^5_{11_1} + k_{1_2} + k_{2_2_2} + k_{3_2_2} = 0 \\
 &y^5_{0_1} + [y^5_{10_1}] + [x^6_{12_1}] + k_{1_{11}} + k_{2_{11_2}} + k_{3_{11_2}} = 0 \\
 &y^5_{1_1} + y^5_{11_1} + y^5_{14_1} + x^6_{1_1} + k_{1_{13}} + k_{2_{13_2}} + k_{3_{13_2}} = 0 \\
 &[y^5_{6_1}] + [y^5_{9_1}] + x^6_{11_1} + k_{1_8} + k_{2_8_2} + k_{3_8_2} = 0 \\
 &y^6_{1_1} + y^6_{11_1} + [y^6_{14_1}] + [x^7_{1_1}] + k_{1_5} + k_{2_5_3} + k_{3_5_3} = 0 \\
 &S(x^3_{5_1}) + y^3_{5_1} = 0 \\
 &S(x^3_{13_1}) + y^3_{13_1} = 0 \\
 &S(x^4_{6_1}) + y^4_{6_1} = 0 \\
 &S(x^4_{10_1}) + y^4_{10_1} = 0 \\
 &S(x^4_{14_1}) + y^4_{14_1} = 0 \\
 &S(x^5_{0_1}) + y^5_{0_1} = 0 \\
 &S(x^5_{1_1}) + y^5_{1_1} = 0 \\
 &S(x^5_{11_1}) + y^5_{11_1} = 0 \\
 &S(x^5_{14_1}) + y^5_{14_1} = 0 \\
 &S(x^6_{1_1}) + y^6_{1_1} = 0 \\
 &S(x^6_{11_1}) + y^6_{11_1} = 0
 \end{aligned}$$

the 1 th Minimal Constraint set is:

$$\begin{aligned}
 &y^1_{1_1} + x^2_{5_1} + k_{1_{15}} + k_{2_{15_0}} + k_{3_{15_0}} = 0 \\
 &y^1_{1_1} + y^1_{11_1} + x^2_{13_1} + k_{1_{15}} + k_{2_{15_0}} + k_{3_{15_0}} = 0 \\
 &y^1_{1_3} + y^1_{9_1} + [y^1_{12_1}] + [x^2_{3_1}] + k_{1_{13}} + k_{2_{13_0}} + k_{3_{13_0}} = 0 \\
 &y^1_{1_3} + y^1_{9_1} + x^2_{15_1} + k_{1_{13}} + k_{2_{13_0}} + k_{3_{13_0}} = 0 \\
 &[y^1_{4_1}] + y^1_{11_1} + [x^2_{9_1}] + k_{1_{10}} + k_{2_{10_0}} + k_{3_{10_0}} = 0 \\
 &[y^2_{0_1}] + y^2_{10_1} + y^2_{13_1} + [x^3_{0_1}] + k_{1_1} + k_{2_1_1} + k_{3_1_1} = 0 \\
 &[y^2_{0_1}] + y^2_{10_1} + [x^3_{12_1}] + k_{1_1} + k_{2_1_1} + k_{3_1_1} = 0 \\
 &y^2_{2_1} + y^2_{8_1} + y^2_{15_1} + x^3_{2_1} + k_{1_0} + k_{2_0_1} + k_{3_0_1} = 0 \\
 &y^2_{2_1} + x^3_{6_1} + k_{1_0} + k_{2_0_1} + k_{3_0_1} = 0 \\
 &y^2_{5_1} + y^2_{8_1} + x^3_{10_1} + k_{1_6} + k_{2_6_1} + k_{3_6_1} = 0 \\
 &y^3_{2_1} + x^4_{6_1} + k_{1_9} + k_{2_9_1} + k_{3_9_1} = 0 \\
 &y^3_{6_1} + [y^3_{9_1}] + [x^4_{11_1}] + k_{1_{10}} + k_{2_{10_1}} + k_{3_{10_1}} = 0
 \end{aligned}$$

$$\begin{aligned}
&[y^3_7] + y^3_{10} + [x^4_8] + k_{1,13} + k_{2,13,1} + k_{3,13,1} = 0 \\
&[y^4_3] + y^4_9 + [x^5_{15}] + k_{1,6} + k_{2,6,2} + k_{3,6,2} = 0 \\
&y^4_6 + y^4_9 + x^5_{11} + k_{1,2} + k_{2,2,2} + k_{3,2,2} = 0 \\
&[y^5_4] + y^5_{11} + [x^6_9] + k_{1,9} + k_{2,9,2} + k_{3,9,2} = 0 \\
&S(x^2_5) + y^2_5 = 0 \\
&S(x^2_{13}) + y^2_{13} = 0 \\
&S(x^2_{15}) + y^2_{15} = 0 \\
&S(x^3_2) + y^3_2 = 0 \\
&S(x^3_6) + y^3_6 = 0 \\
&S(x^3_{10}) + y^3_{10} = 0 \\
&S(x^4_6) + y^4_6 = 0 \\
&S(x^5_{11}) + y^5_{11} = 0
\end{aligned}$$

the 2 th Minimal Constraint set is:

$$\begin{aligned}
&y^1_1 + x^2_5 + k_{1,15} + k_{2,15,0} + k_{3,15,0} = 0 \\
&y^1_1 + y^1_{11} + x^2_{13} + k_{1,15} + k_{2,15,0} + k_{3,15,0} = 0 \\
&y^1_3 + y^1_9 + [y^1_{12}] + [x^2_3] + k_{1,13} + k_{2,13,0} + k_{3,13,0} = 0 \\
&y^1_3 + y^1_9 + x^2_{15} + k_{1,13} + k_{2,13,0} + k_{3,13,0} = 0 \\
&[y^1_4] + y^1_{11} + [x^2_9] + k_{1,10} + k_{2,10,0} + k_{3,10,0} = 0 \\
&[y^2_0] + y^2_{10} + y^2_{13} + [x^3_0] + k_{1,1} + k_{2,1,1} + k_{3,1,1} = 0 \\
&[y^2_0] + y^2_{10} + [x^3_{12}] + k_{1,1} + k_{2,1,1} + k_{3,1,1} = 0 \\
&y^2_1 + x^3_5 + k_{1,7} + k_{2,7,1} + k_{3,7,1} = 0 \\
&y^2_1 + y^2_{11} + x^3_{13} + k_{1,7} + k_{2,7,1} + k_{3,7,1} = 0 \\
&y^2_2 + y^2_8 + y^2_{15} + x^3_2 + k_{1,0} + k_{2,0,1} + k_{3,0,1} = 0 \\
&y^2_2 + x^3_6 + k_{1,0} + k_{2,0,1} + k_{3,0,1} = 0 \\
&[y^2_3] + [y^2_9] + [y^2_{12}] + x^3_3 + k_{1,5} + k_{2,5,1} + k_{3,5,1} = 0 \\
&[y^2_4] + y^2_{11} + [x^3_9] + k_{1,2} + k_{2,2,1} + k_{3,2,1} = 0 \\
&y^2_5 + y^2_8 + x^3_{10} + k_{1,6} + k_{2,6,1} + k_{3,6,1} = 0 \\
&[y^3_0] + y^3_{10} + y^3_{13} + [x^4_0] + k_{1,15} + k_{2,15,1} + k_{3,15,1} = 0 \\
&y^3_2 + y^3_8 + x^4_{14} + k_{1,9} + k_{2,9,1} + k_{3,9,1} = 0 \\
&y^3_3 + x^4_7 + k_{1,14} + k_{2,14,1} + k_{3,14,1} = 0 \\
&y^3_5 + y^3_8 + x^4_{10} + k_{1,12} + k_{2,12,1} + k_{3,12,1} = 0 \\
&y^3_6 + [y^3_9] + [x^4_{11}] + k_{1,10} + k_{2,10,1} + k_{3,10,1} = 0 \\
&[y^4_1] + [y^4_{11}] + y^4_{14} + x^5_1 + k_{1,3} + k_{2,3,2} + k_{3,3,2} = 0 \\
&y^4_7 + y^4_{10} + x^5_8 + k_{1,5} + k_{2,5,2} + k_{3,5,2} = 0 \\
&y^5_1 + x^6_5 + k_{1,13} + k_{2,13,2} + k_{3,13,2} = 0 \\
&[y^5_2] + y^5_8 + [y^5_{15}] + x^6_2 + k_{1,15} + k_{2,15,2} + k_{3,15,2} = 0 \\
&[y^5_3] + [y^5_9] + x^6_{15} + k_{1,12} + k_{2,12,2} + k_{3,12,2} = 0 \\
&y^6_2 + y^6_8 + y^6_{15} + x^7_2 + k_{1,7} + k_{2,7,3} + k_{3,7,3} = 0 \\
&y^6_5 + y^6_8 + x^7_{10} + k_{1,2} + k_{2,2,3} + k_{3,2,3} = 0 \\
&[y^6_7] + [y^6_{10}] + x^7_8 + k_{1,6} + k_{2,6,3} + k_{3,6,3} = 0 \\
&y^7_2 + y^7_8 + [y^7_{15}] + [x^8_2] + k_{1,11} + k_{2,11,3} + k_{3,11,3} = 0 \\
&[y^7_7] + y^7_{10} + [x^8_8] + k_{1,12} + k_{2,12,3} + k_{3,12,3} = 0 \\
&S(x^2_5) + y^2_5 = 0 \\
&S(x^2_{13}) + y^2_{13} = 0
\end{aligned}$$

$$S(x^{2}_{15}) + y^{2}_{15} = 0$$

$$S(x^{3}_{2}) + y^{3}_{2} = 0$$

$$S(x^{3}_{3}) + y^{3}_{3} = 0$$

$$S(x^{3}_{5}) + y^{3}_{5} = 0$$

$$S(x^{3}_{6}) + y^{3}_{6} = 0$$

$$S(x^{3}_{10}) + y^{3}_{10} = 0$$

$$S(x^{3}_{13}) + y^{3}_{13} = 0$$

$$S(x^{4}_{7}) + y^{4}_{7} = 0$$

$$S(x^{4}_{10}) + y^{4}_{10} = 0$$

$$S(x^{4}_{14}) + y^{4}_{14} = 0$$

$$S(x^{5}_{1}) + y^{5}_{1} = 0$$

$$S(x^{5}_{8}) + y^{5}_{8} = 0$$

$$S(x^{6}_{2}) + y^{6}_{2} = 0$$

$$S(x^{6}_{5}) + y^{6}_{5} = 0$$

$$S(x^{6}_{15}) + y^{6}_{15} = 0$$

$$S(x^{7}_{2}) + y^{7}_{2} = 0$$

$$S(x^{7}_{8}) + y^{7}_{8} = 0$$

$$S(x^{7}_{10}) + y^{7}_{10} = 0$$

the 3 th Minimal Constraint set is:

$$[y^{2}_{3}] + [y^{2}_{9}] + [y^{2}_{12}] + x^{3}_{3} + k1_{5} + k2_{5_1} + k3_{5_1} = 0$$

$$y^{3}_{3} + x^{4}_{7} + k1_{14} + k2_{14_1} + k3_{14_1} = 0$$

$$[y^{4}_{0}] + y^{4}_{10} + [y^{4}_{13}] + x^{5}_{0} + k1_{7} + k2_{7_2} + k3_{7_2} = 0$$

$$y^{4}_{7} + y^{4}_{10} + x^{5}_{8} + k1_{5} + k2_{5_2} + k3_{5_2} = 0$$

$$y^{5}_{0} + [y^{5}_{10}] + [x^{6}_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0$$

$$[y^{5}_{2}] + y^{5}_{8} + [y^{5}_{15}] + x^{6}_{2} + k1_{15} + k2_{15_2} + k3_{15_2} = 0$$

$$[y^{5}_{3}] + [y^{5}_{9}] + x^{6}_{15} + k1_{12} + k2_{12_2} + k3_{12_2} = 0$$

$$[y^{5}_{7}] + [y^{5}_{10}] + x^{6}_{8} + k1_{14} + k2_{14_2} + k3_{14_2} = 0$$

$$y^{6}_{2} + y^{6}_{8} + y^{6}_{15} + x^{7}_{2} + k1_{7} + k2_{7_3} + k3_{7_3} = 0$$

$$[y^{6}_{7}] + [y^{6}_{10}] + x^{7}_{8} + k1_{6} + k2_{6_3} + k3_{6_3} = 0$$

$$y^{7}_{2} + y^{7}_{8} + [y^{7}_{15}] + [x^{8}_{2}] + k1_{11} + k2_{11_3} + k3_{11_3} = 0$$

$$S(x^{3}_{3}) + y^{3}_{3} = 0$$

$$S(x^{4}_{7}) + y^{4}_{7} = 0$$

$$S(x^{5}_{0}) + y^{5}_{0} = 0$$

$$S(x^{5}_{8}) + y^{5}_{8} = 0$$

$$S(x^{6}_{2}) + y^{6}_{2} = 0$$

$$S(x^{6}_{8}) + y^{6}_{8} = 0$$

$$S(x^{6}_{15}) + y^{6}_{15} = 0$$

$$S(x^{7}_{2}) + y^{7}_{2} = 0$$

$$S(x^{7}_{8}) + y^{7}_{8} = 0$$

the 4 th Minimal Constraint set is:

$$[y^{4}_{1}] + [y^{4}_{11}] + x^{5}_{13} + k1_{3} + k2_{3_2} + k3_{3_2} = 0$$

$$y^{5}_{0} + [y^{5}_{10}] + y^{5}_{13} + [x^{6}_{0}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0$$

$$y^{5}_{0} + [y^{5}_{10}] + [x^{6}_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0$$

$$S(x^{5}_{13}) + y^{5}_{13} = 0$$

the 5 th Minimal Constraint set is:

$$\begin{aligned}
& [y^2_3] + [y^2_9] + [y^2_{12}] + x^3_3 + k1_5 + k2_{5_1} + k3_{5_1} = 0 \\
& y^3_3 + x^4_7 + k1_{14} + k2_{14_1} + k3_{14_1} = 0 \\
& [y^4_0] + y^4_{10} + [y^4_{13}] + x^5_0 + k1_7 + k2_{7_2} + k3_{7_2} = 0 \\
& y^4_7 + y^4_{10} + x^5_8 + k1_5 + k2_{5_2} + k3_{5_2} = 0 \\
& y^5_0 + [y^5_{10}] + [x^6_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0 \\
& [y^5_5] + y^5_8 + [x^6_{10}] + k1_{10} + k2_{10_2} + k3_{10_2} = 0 \\
& S(x^3_3) + y^3_3 = 0 \\
& S(x^4_7) + y^4_7 = 0 \\
& S(x^5_0) + y^5_0 = 0 \\
& S(x^5_8) + y^5_8 = 0
\end{aligned}$$

the 6 th Minimal Constraint set is:

$$\begin{aligned}
& [y^2_3] + [y^2_9] + [y^2_{12}] + x^3_3 + k1_5 + k2_{5_1} + k3_{5_1} = 0 \\
& y^3_3 + x^4_7 + k1_{14} + k2_{14_1} + k3_{14_1} = 0 \\
& [y^4_0] + y^4_{10} + [y^4_{13}] + x^5_0 + k1_7 + k2_{7_2} + k3_{7_2} = 0 \\
& y^4_7 + y^4_{10} + x^5_8 + k1_5 + k2_{5_2} + k3_{5_2} = 0 \\
& y^5_0 + [y^5_{10}] + [x^6_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0 \\
& [y^5_2] + y^5_8 + [x^6_{14}] + k1_{15} + k2_{15_2} + k3_{15_2} = 0 \\
& S(x^3_3) + y^3_3 = 0 \\
& S(x^4_7) + y^4_7 = 0 \\
& S(x^5_0) + y^5_0 = 0 \\
& S(x^5_8) + y^5_8 = 0
\end{aligned}$$

the 7 th Minimal Constraint set is:

$$\begin{aligned}
& [y^4_0] + y^4_{10} + [y^4_{13}] + x^5_0 + k1_7 + k2_{7_2} + k3_{7_2} = 0 \\
& [y^4_0] + y^4_{10} + [x^5_{12}] + k1_7 + k2_{7_2} + k3_{7_2} = 0 \\
& y^5_0 + [y^5_{10}] + [x^6_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0 \\
& S(x^5_0) + y^5_0 = 0
\end{aligned}$$

the 8 th Minimal Constraint set is:

$$\begin{aligned}
& y^1_1 + x^2_5 + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
& y^1_1 + y^1_{11} + x^2_{13} + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
& y^1_3 + y^1_9 + [y^1_{12}] + [x^2_3] + k1_{13} + k2_{13_0} + k3_{13_0} = 0 \\
& y^1_3 + y^1_9 + x^2_{15} + k1_{13} + k2_{13_0} + k3_{13_0} = 0 \\
& [y^1_4] + y^1_{11} + [x^2_9] + k1_{10} + k2_{10_0} + k3_{10_0} = 0 \\
& [y^2_0] + y^2_{10} + y^2_{13} + [x^3_0] + k1_1 + k2_{1_1} + k3_{1_1} = 0 \\
& [y^2_0] + y^2_{10} + [x^3_{12}] + k1_1 + k2_{1_1} + k3_{1_1} = 0 \\
& y^2_1 + x^3_5 + k1_7 + k2_{7_1} + k3_{7_1} = 0 \\
& y^2_1 + y^2_{11} + x^3_{13} + k1_7 + k2_{7_1} + k3_{7_1} = 0 \\
& y^2_2 + y^2_8 + y^2_{15} + x^3_2 + k1_0 + k2_{0_1} + k3_{0_1} = 0 \\
& y^2_2 + x^3_6 + k1_0 + k2_{0_1} + k3_{0_1} = 0 \\
& [y^2_4] + y^2_{11} + [x^3_9] + k1_2 + k2_{2_1} + k3_{2_1} = 0 \\
& y^2_5 + y^2_8 + x^3_{10} + k1_6 + k2_{6_1} + k3_{6_1} = 0 \\
& [y^3_0] + y^3_{10} + y^3_{13} + [x^4_0] + k1_{15} + k2_{15_1} + k3_{15_1} = 0 \\
& y^3_2 + y^3_8 + [y^3_{15}] + [x^4_2] + k1_9 + k2_{9_1} + k3_{9_1} = 0 \\
& y^3_5 + y^3_8 + x^4_{10} + k1_{12} + k2_{12_1} + k3_{12_1} = 0
\end{aligned}$$

$$\begin{aligned}
&y^3_6 + [y^3_9] + [x^4_{11}] + k1_{10} + k2_{10_1} + k3_{10_1} = 0 \\
&[y^4_0] + y^4_{10} + [y^4_{13}] + x^5_0 + k1_7 + k2_{7_2} + k3_{7_2} = 0 \\
&y^5_0 + [y^5_{10}] + [x^6_{12}] + k1_{11} + k2_{11_2} + k3_{11_2} = 0 \\
&S(x^2_5) + y^2_5 = 0 \\
&S(x^2_{13}) + y^2_{13} = 0 \\
&S(x^2_{15}) + y^2_{15} = 0 \\
&S(x^3_2) + y^3_2 = 0 \\
&S(x^3_5) + y^3_5 = 0 \\
&S(x^3_6) + y^3_6 = 0 \\
&S(x^3_{10}) + y^3_{10} = 0 \\
&S(x^3_{13}) + y^3_{13} = 0 \\
&S(x^4_{10}) + y^4_{10} = 0 \\
&S(x^5_0) + y^5_0 = 0
\end{aligned}$$

the 9 th Minimal Constraint set is:

$$\begin{aligned}
&[y^3_4] + [y^3_{11}] + x^4_9 + k1_8 + k2_{8_1} + k3_{8_1} = 0 \\
&[y^4_3] + y^4_9 + [x^5_{15}] + k1_6 + k2_{6_2} + k3_{6_2} = 0 \\
&S(x^4_9) + y^4_9 = 0
\end{aligned}$$

the 10 th Minimal Constraint set is:

$$\begin{aligned}
&[y^4_3] + y^4_9 + [y^4_{12}] + [x^5_3] + k1_6 + k2_{6_2} + k3_{6_2} = 0 \\
&[y^4_3] + y^4_9 + [x^5_{15}] + k1_6 + k2_{6_2} + k3_{6_2} = 0
\end{aligned}$$

the 11 th Minimal Constraint set is:

$$\begin{aligned}
&y^1_1 + x^2_5 + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
&y^1_1 + y^1_{11} + x^2_{13} + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
&[y^1_4] + y^1_{11} + [x^2_9] + k1_{10} + k2_{10_0} + k3_{10_0} = 0 \\
&[y^2_0] + y^2_{10} + y^2_{13} + [x^3_0] + k1_1 + k2_{1_1} + k3_{1_1} = 0 \\
&[y^2_0] + y^2_{10} + [x^3_{12}] + k1_1 + k2_{1_1} + k3_{1_1} = 0 \\
&y^2_2 + x^3_6 + k1_0 + k2_{0_1} + k3_{0_1} = 0 \\
&y^2_2 + y^2_8 + x^3_{14} + k1_0 + k2_{0_1} + k3_{0_1} = 0 \\
&y^2_5 + y^2_8 + x^3_{10} + k1_6 + k2_{6_1} + k3_{6_1} = 0 \\
&y^3_1 + [y^3_{11}] + y^3_{14} + [x^4_1] + k1_{11} + k2_{11_1} + k3_{11_1} = 0 \\
&y^3_1 + x^4_5 + k1_{11} + k2_{11_1} + k3_{11_1} = 0 \\
&y^3_6 + [y^3_9] + [x^4_{11}] + k1_{10} + k2_{10_1} + k3_{10_1} = 0 \\
&[y^3_7] + y^3_{10} + [x^4_8] + k1_{13} + k2_{13_1} + k3_{13_1} = 0 \\
&y^4_5 + [y^4_8] + [x^5_{10}] + k1_4 + k2_{4_2} + k3_{4_2} = 0 \\
&S(x^2_5) + y^2_5 = 0 \\
&S(x^2_{13}) + y^2_{13} = 0 \\
&S(x^3_6) + y^3_6 = 0 \\
&S(x^3_{10}) + y^3_{10} = 0 \\
&S(x^3_{14}) + y^3_{14} = 0 \\
&S(x^4_5) + y^4_5 = 0
\end{aligned}$$

the 12 th Minimal Constraint set is:

$$\begin{aligned}
&y^1_1 + x^2_5 + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
&y^1_1 + y^1_{11} + x^2_{13} + k1_{15} + k2_{15_0} + k3_{15_0} = 0 \\
&[y^1_4] + y^1_{11} + [x^2_9] + k1_{10} + k2_{10_0} + k3_{10_0} = 0
\end{aligned}$$

$$\begin{aligned}
&[y^2_0] + y^2_{10} + y^2_{13} + [x^3_0] + k_{1_1} + k_{2_1_1} + k_{3_1_1} = 0 \\
&[y^2_0] + y^2_{10} + [x^3_{12}] + k_{1_1} + k_{2_1_1} + k_{3_1_1} = 0 \\
&y^2_2 + x^3_6 + k_{1_0} + k_{2_0_1} + k_{3_0_1} = 0 \\
&y^2_2 + y^2_8 + x^3_{14} + k_{1_0} + k_{2_0_1} + k_{3_0_1} = 0 \\
&y^2_5 + y^2_8 + x^3_{10} + k_{1_6} + k_{2_6_1} + k_{3_6_1} = 0 \\
&y^3_1 + [y^3_{11}] + y^3_{14} + [x^4_1] + k_{1_{11}} + k_{2_{11_1}} + k_{3_{11_1}} = 0 \\
&y^3_1 + [y^3_{11}] + [x^4_{13}] + k_{1_{11}} + k_{2_{11_1}} + k_{3_{11_1}} = 0 \\
&y^3_6 + [y^3_9] + [x^4_{11}] + k_{1_{10}} + k_{2_{10_1}} + k_{3_{10_1}} = 0 \\
&[y^3_7] + y^3_{10} + [x^4_8] + k_{1_{13}} + k_{2_{13_1}} + k_{3_{13_1}} = 0 \\
&S(x^2_5) + y^2_5 = 0 \\
&S(x^2_{13}) + y^2_{13} = 0 \\
&S(x^3_6) + y^3_6 = 0 \\
&S(x^3_{10}) + y^3_{10} = 0 \\
&S(x^3_{14}) + y^3_{14} = 0
\end{aligned}$$

the 13 th Minimal Constraint set is:

$$\begin{aligned}
&[y^3_0] + y^3_{10} + [x^4_{12}] + k_{1_{15}} + k_{2_{15_1}} + k_{3_{15_1}} = 0 \\
&[y^3_7] + y^3_{10} + [x^4_8] + k_{1_{13}} + k_{2_{13_1}} + k_{3_{13_1}} = 0
\end{aligned}$$

the 14 th Minimal Constraint set is:

$$\begin{aligned}
&[y^2_3] + [y^2_9] + [y^2_{12}] + x^3_3 + k_{1_5} + k_{2_{5_1}} + k_{3_{5_1}} = 0 \\
&y^3_3 + [y^3_9] + [x^4_{15}] + k_{1_{14}} + k_{2_{14_1}} + k_{3_{14_1}} = 0 \\
&S(x^3_3) + y^3_3 = 0
\end{aligned}$$

the 15 th Minimal Constraint set is:

$$\begin{aligned}
&[y^2_3] + [y^2_9] + [y^2_{12}] + x^3_3 + k_{1_5} + k_{2_{5_1}} + k_{3_{5_1}} = 0 \\
&y^3_3 + [y^3_9] + [y^3_{12}] + [x^4_3] + k_{1_{14}} + k_{2_{14_1}} + k_{3_{14_1}} = 0 \\
&S(x^3_3) + y^3_3 = 0
\end{aligned}$$

the 16 th Minimal Constraint set is:

$$[y^0_0] + [x^1_4] + k_{1_0} + k_{2_{0_0}} + k_{3_{0_0}} = 0$$

the 17 th Minimal Constraint set is:

$$[y^1_0] + [x^2_4] + k_{1_9} + k_{2_{9_0}} + k_{3_{9_0}} = 0$$

the 18 th Minimal Constraint set is:

$$[y^2_0] + [x^3_4] + k_{1_1} + k_{2_{1_1}} + k_{3_{1_1}} = 0$$

the 19 th Minimal Constraint set is:

$$[y^2_3] + [x^3_7] + k_{1_5} + k_{2_{5_1}} + k_{3_{5_1}} = 0$$

the 20 th Minimal Constraint set is:

$$[y^2_3] + [y^2_9] + [x^3_{15}] + k_{1_5} + k_{2_{5_1}} + k_{3_{5_1}} = 0$$

the 21 th Minimal Constraint set is:

$$[y^3_0] + [x^4_4] + k_{1_{15}} + k_{2_{15_1}} + k_{3_{15_1}} = 0$$

the 22 th Minimal Constraint set is:

$$[y^4_0] + [x^5_4] + k_{1_7} + k_{2_{7_2}} + k_{3_{7_2}} = 0$$

the 23 th Minimal Constraint set is:

$$[y^4_1] + [x^5_5] + k_{1_3} + k_{2_{3_2}} + k_{3_{3_2}} = 0$$

the 24 th Minimal Constraint set is:

$$[y^4_2] + [y^4_8] + [y^4_{15}] + [x^5_2] + k_{1_1} + k_{2_{1_2}} + k_{3_{1_2}} = 0$$

the 25 th Minimal Constraint set is:

$$[y^4_2] + [x^5_6] + k1_1 + k2_1_2 + k3_1_2 = 0$$

the 26 th Minimal Constraint set is:

$$[y^4_3] + [x^5_7] + k1_6 + k2_6_2 + k3_6_2 = 0$$

the 27 th Minimal Constraint set is:

$$[y^4_4] + [y^4_{11}] + [x^5_9] + k1_0 + k2_0_2 + k3_0_2 = 0$$

the 28 th Minimal Constraint set is:

$$[y^5_2] + [x^6_6] + k1_{15} + k2_{15_2} + k3_{15_2} = 0$$

the 29 th Minimal Constraint set is:

$$[y^5_3] + [y^5_9] + [y^5_{12}] + [x^6_3] + k1_{12} + k2_{12_2} + k3_{12_2} = 0$$

the 30 th Minimal Constraint set is:

$$[y^5_3] + [x^6_7] + k1_{12} + k2_{12_2} + k3_{12_2} = 0$$

the 31 th Minimal Constraint set is:

$$[y^6_0] + [x^7_4] + k1_3 + k2_3_3 + k3_3_3 = 0$$

the 32 th Minimal Constraint set is:

$$[y^6_3] + [x^7_7] + k1_4 + k2_4_3 + k3_4_3 = 0$$

the 33 th Minimal Constraint set is:

$$[y^6_3] + [y^6_9] + [x^7_{15}] + k1_4 + k2_4_3 + k3_4_3 = 0$$

the 34 th Minimal Constraint set is:

$$[y^6_6] + [y^6_9] + [x^7_{11}] + k1_0 + k2_0_3 + k3_0_3 = 0$$

the 35 th Minimal Constraint set is:

$$[y^7_1] + [x^8_5] + k1_{14} + k2_{14_3} + k3_{14_3} = 0$$

the 36 th Minimal Constraint set is:

$$[y^8_2] + [x^9_6] + k1_3 + k2_3_4 + k3_3_4 = 0$$

the 37 th Minimal Constraint set is:

$$S([x^0_0]) + [y^0_0] = 0$$